

Design, Modeling, And Experimental Evaluation Of A Modular Magnetically Coupled Robotic Gripper

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Abstract—Reconfigurable end effectors are essential for high-mix automation but are often limited by increased distal inertia and wrist cable management. We present a modular robotic gripper using a passive permanent-magnet coupling to enable fast, low-inertia tool exchange without mechanical locks or electrical connectors. The design employs decentralized, underactuated finger modules driven by embedded servos, allowing grasp geometry and compliance to be changed independently of the palm kinematics. A hybrid modeling pipeline characterizes the magnetic interface via finite-element magnetostatic analysis to estimate flux distribution and shear capacity. Those results are embedded in a ROS–Gazebo simulation through a custom plugin that models magnetic restoring forces together with servo dynamics; domain randomization of contact and actuation parameters is used to reduce sim-to-real sensitivity. A sensorized prototype is validated on the YCB benchmark, demonstrating docking repeatability within ± 0.5 mm, grasp success rates $>90\%$ within the payload range, and tool exchange times below 1.0 s. These findings show that properly modeled passive magnetic coupling can provide a scalable, low-inertia solution for agile modular manipulation.

Keywords: Modular robotic gripper; magnetic coupling; reconfigurable end effectors; underactuated mechanisms; sim-to-real transfer; YCB benchmark.